

Statistical methods for computer science

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Introduction to Random Variables (Part I)

Where were we?

Statistical science is the science of developing and applying methods for collecting, analyzing and interpreting data.

- ✓ Design phase
 - ✓ Description phase
 - ✓ Probability
 - Introduction to random variables
- Today** Discrete random variables
- Today** Continuous random variables

Course Outline

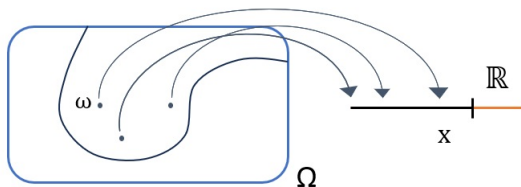
- Basic concepts
- Descriptive statistics (+ R Lab)
- Introduction to probability
- **Introduction to random variables**
- Discrete probability distributions
- Continuous probability distributions (+ R Lab)
- Sampling distributions
- Statistical inference: estimation (+ R Lab)
- Statistical inference: testing (+ R Lab)
- Linear model: estimation, inference, diagnostics (+ R Lab)

Random Variable

Given a probability space $(\Omega, \mathcal{A}, P)$, a **random variable** is a measurable function mapping sample points to numbers $X : \Omega \longrightarrow \mathbb{R}$ such that

$$\forall x \in \mathbb{R}, \{\omega : X(\omega) \leq x\} \in \mathcal{A} \quad (1)$$

The measurability condition (1) is a technical condition needed to transfer probability, defined on subsets of Ω , to subsets of $\mathbb{R}$, i.e.,
 $P(X \leq x) = P(\omega : X(\omega) \leq x)$



set of the ω 's such that $X(\omega) \leq x$. By assumption it belongs to $\mathcal{A}$, so it has a probability

Types of random variables

The set of values taken by a RV is called *domain*. Based on that, we can distinguish between:

Discrete RVs

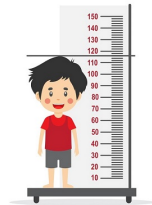
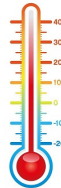
take on a countable number of distinct values



Fully described by:
Cumulative Distribution Function (CDF)
Probability Mass Function (PMF)

Continuous RVs

take on uncountably many values



Fully described by:
Cumulative Distribution Function (CDF)
Probability Density Function (PDF)

Discrete random variables

Discrete RVs

take on a countable number of distinct values



Given the probability space $(\Omega, \mathcal{A}, P)$ and a discrete set $\mathcal{X} = \{x_1, x_2, \dots\} \subset \mathbb{R}$ a discrete random variable is a function $X : \Omega \rightarrow \mathcal{X}$ such that $\forall x \in \mathcal{X}$
 $\{\omega : X(\omega) = x\} \in \mathcal{A}$

Fully described by:
Cumulative Distribution Function
(CDF)
Probability Mass Function
(PMF)

- PDF
- CDF
- Expectation
- Variance

Probability Mass Function

Let X be a discrete random variable. The PMF of X is a function $f_X : \mathcal{X} \rightarrow \mathbb{R}$ that assigns probabilities to each possible realization of X

$$f_X(x) = P(X = x) \quad \forall x \in \mathbb{R}$$

Note: the notation stands for $\underbrace{f_X}_{R.V.}(\underbrace{x}_{\text{realization}}) = P(\underbrace{X}_{R.V.} = \underbrace{x}_{\text{realization}})$

Properties:

- $f_X(x) \geq 0$
- $\sum_{x \in \mathcal{D}_f} f_X(x) = 1$

Example

Let's consider the random experiment “toss 2 fair coins” and define the random variable X that counts the number of heads.

- Domain:

Example

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- Domain: $\mathcal{D}_f = \{0, 1, 2\}$
- PMF :

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- Domain: $\mathcal{D}_f = \{0, 1, 2\}$

- PMF : $f_X(x) = \begin{cases} 1/4 & \text{if } x = 0 \\ 1/2 & \text{if } x = 1 \\ 1/4 & \text{if } x = 2 \\ 0 & \text{otherwise} \end{cases}$

x	$f_X(x)$
0	0.25
1	0.50
2	0.25

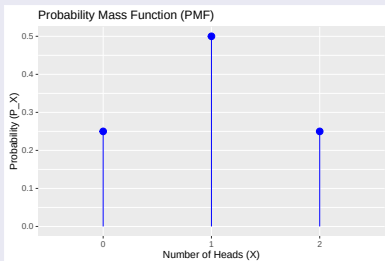
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x	$f_X(x)$
0	0.25
1	0.50
2	0.25



- ✓ $f_X(x) \geq 0 \quad \forall x$
- ✓ $\sum_{x \in \mathcal{D}_f} f_X(x) = 1$

Cumulative Distribution Function

Let X be a discrete random variable. The CDF of X is a function $F_X : \mathcal{X} \rightarrow \mathbb{R}$ given by

$$F_X(x) = P(X \leq x) \quad \forall x \in \mathbb{R}$$

Note: the notation stands for $\underbrace{F_X}_{R.V.}(\underbrace{x}_{realization}) = P(\underbrace{X}_{R.V.} \leq \underbrace{x}_{realization})$

Properties:

- $0 \leq F_X(x) \leq 1$
- $\lim_{x \rightarrow -\infty} F_X(x) = 0, \lim_{x \rightarrow +\infty} F_X(x) = 1$
- $F_X(x)$ is non-decreasing

Example

Let's consider the random experiment “toss 2 fair coins” and define the random variable X that counts the number of heads.

- Domain: $\mathcal{D}_f = \{0, 1, 2\}$
- CDF :

Example

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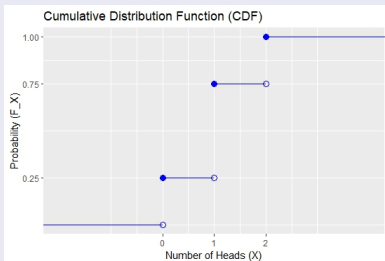
- Domain: $\mathcal{D}_f = \{0, 1, 2\}$
- CDF : $F_X(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1/4 & \text{if } 0 \leq x < 1 \\ 3/4 & \text{if } 1 \leq x < 2 \\ 1 & \text{if } x \geq 2 \end{cases}$

Example

Let's consider the random experiment "toss 2 fair coins" and define the random variable X that counts the number of heads.

- Domain: $\mathcal{D}_f = \{0, 1, 2\}$

- CDF : $F_X(x) = \begin{cases} 0 & \text{if } x < 0 \\ 1/4 & \text{if } 0 \leq x < 1 \\ 3/4 & \text{if } 1 \leq x < 2 \\ 1 & \text{if } x \geq 2 \end{cases}$



- ✓ $0 \leq F_X(x) \leq 1$
- ✓ $\lim_{x \rightarrow -\infty} F_X(x) = 0,$
 $\lim_{x \rightarrow +\infty} F_X(x) = 1$
- ✓ Non-decreasing

Link between PMF and CDF

How are the PMF and the CDF related?

- The CDF can be obtained from the PMF by summing up the PMF values up to and including x

$$F_X(x) = P(X \leq x) = \sum_{x_i \leq x} f_X(x_i)$$

- The PMF can be obtained from the CDF by taking the difference between the CDF values at two consecutive points of the random variable

$$f_X(x) = P(X = x) = F_X(x) - \lim_{t \rightarrow x^-} F_X(t)$$

The PMF and the CDF fully describe a random variable, but sometimes we may be interested in something less informative and easier to interpret

Expectation

Let X be a discrete random variable whose PMF is given by $f_X(x)$. The *expectation* or *mean* of X is given by

$$E[X] = \mu = \sum_x x \cdot f_X(x) \quad (2)$$

More generally, we can define expectations for functions of random variables.

The expectation of a real-valued function $g(X)$ of a discrete random variable X with PMF given by $f_X(x)$, is

$$E[g(X)] = \sum_x g(x) \cdot f_X(x) \quad (3)$$

Expectation

Special functions $g(X)$:

$g(X)$	$E[g(X)]$	
$g(X) = X$	$E[g(X)] = \sum_x x f_X(x) = \mu$	Equation (3) reduces to (2), the mean of X
$g(X) = X^2$	$E[g(X)] = \sum_x x^2 f_X(x) = \mu_2$	Second moment of X
$g(X) = (X - \mu)^2$	$E[g(X)] = \sum_x (x - \mu)^2 f_X(x) = \sigma^2$	Second centered moment of X or variance

Exercise

Let X be a RV with the following CDF

$$F_X(x) = \begin{cases} 0 & \text{if } x < -1 \\ 0.6 & \text{if } -1 \leq x < 0 \\ 0.8 & \text{if } 0 \leq x < 2 \\ 1 & \text{if } x \geq 2 \end{cases}$$

- 1 Is $F_X(x)$ a valid CDF?
- 2 Get the corresponding PMF and draw it
- 3 Compute $E[X]$ and $E[X^2]$
- 4 Compute $\text{VAR}[X]$

Exercise

① Yes, it is a valid CDF because all the properties are satisfied:

- ✓ $0 \leq F_X(x) \leq 1$
- ✓ $\lim_{x \rightarrow -\infty} F_X(x) = 0$, $\lim_{x \rightarrow +\infty} F_X(x) = 1$, $F_X(x)$
- ✓ $F_X(x)$ is non-decreasing, as $0 < 0.6 < 0.8 < 1$

② The PMF can be calculated as follows

x	$f_X(x)$
-1	$F_X(-1) - F_X(x < -1) = 0.6 - 0 = 0.6$
0	$F_X(0) - F_X(-1) = 0.8 - 0.6 = 0.2$
2	$F_X(2) - F_X(0) = 1 - 0.8 = 0.2$

Exercise

- ③ Computing $E[X]$ and $E[X^2]$:

Exercise

- 3 Computing $E[X]$ and $E[X^2]$:

$$\begin{aligned} E[X] = \mu &= \sum_x x \cdot f_X(x) \\ &= -1 \cdot f_X(-1) + 0 \cdot f_X(0) + 2 \cdot f_X(2) = \\ &= -0.6 + 0.4 = -0.2 \end{aligned}$$

$$\begin{aligned} E[X^2] = \mu_2 &= \sum_x x^2 \cdot f_X(x) \\ &= (-1)^2 \cdot f_X(-1) + 0 \cdot f_X(0) + 2^2 \cdot f_X(2) = \\ &= 0.6 + 0.8 = 1.4 \end{aligned}$$

- 4 Computing $\text{VAR}[X]$:

Exercise

- 3 Computing $E[X]$ and $E[X^2]$:

$$\begin{aligned} E[X] &= \mu = \sum_x x \cdot f_X(x) \\ &= -1 \cdot f_X(-1) + 0 \cdot f_X(0) + 2 \cdot f_X(2) = \\ &= -0.6 + 0.4 = -0.2 \end{aligned}$$

$$\begin{aligned} E[X^2] &= \mu_2 = \sum_x x^2 \cdot f_X(x) \\ &= (-1)^2 \cdot f_X(-1) + 0 \cdot f_X(0) + 2^2 \cdot f_X(2) = \\ &= 0.6 + 0.8 = 1.4 \end{aligned}$$

- 4 Computing $\text{VAR}[X]$:

$$\begin{aligned} \text{VAR}[X] &= \sigma^2 = \sum_x (x - \mu)^2 f_X(x) \\ &= (-1 + 0.2)^2 0.6 + (0 + 0.2)^2 0.2 + (2 + 0.2)^2 0.2 = 1.36 \end{aligned}$$

Variance

Let X be a discrete random variable with mean μ and PMF $f_X(x)$. The *variance* of X is given by

$$V[X] = \sigma^2 = E[(X - \mu)^2] = \sum_x (x - \mu)^2 \cdot f_X(x) \quad (4)$$

Remarks:

- μ is a measure of central tendency (it summarizes the position of the distribution of X along the x-axis) and has the same unit of measure of X
- σ^2 is a measure of the variability of X around its mean but it is measured as X^2
- To get a measure of variability with the same unit of measure of X , use the *standard deviation* $\sigma = \sqrt{\sigma^2}$

Properties

1 *Shortcut for the variance:*

$$E[(X - \mu)^2] = E[X^2] - E[X]^2 = \mu_2 - \mu^2$$

Proof.

$$\begin{aligned} E[(X - \mu)^2] &= \sum_x (x - \mu)^2 f_X(x) = \\ &= \sum_x (x^2 - 2x\mu + \mu^2) f_X(x) = \\ &= \underbrace{\sum_x x^2 f_X(x)}_{\mu_2} - 2\mu \underbrace{\sum_x x f_X(x)}_{\mu} + \mu^2 \underbrace{\sum_x f_X(x)}_1 \\ &= \mu_2 - 2\mu^2 + \mu^2 = \mu_2 - \mu^2 \end{aligned}$$

Properties

2 Expectation of linear transformations of X :

$$E[(a + bX)] = a + bE[X]$$

$$V[(a + bX)] = b^2 V[X]$$

Proof:

$$E[(a + bX)] = \sum_x (a + bx) f_X(x) = a + b \sum_x x \cdot f_X(x)$$

$$= a + bE[X] = a + b\mu$$

$$V[(a + bX)] = \sum_x (a + bx - a - b\mu)^2 f_X(x)$$

$$= \sum_x b^2 (x - \mu)^2 f_X(x)$$

$$= b^2 V[X] = b^2 \sigma^2$$

Properties

3 *Expectation of constants:*

$$E[c] = c$$

$$V[c] = 0$$

Proof: Same as the proof of property 2 with $b = 0$

4 *Expectation of demeaned X :*

$$E[(X - \mu)] = 0$$

$$V[(X - \mu)] = V[X]$$

Proof: Same as the proof of property 2 with $b = 1$ and $a = -\mu$

Properties

4 Standardization of X :

$$E \left[\frac{(X - \mu)}{\sigma} \right] = 0$$

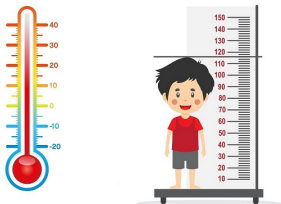
$$V \left[\frac{(X - \mu)}{\sigma} \right] = 1$$

Proof: Same as the proof of property 2 with $b = 1/\sigma$ and $a = -\mu/\sigma$

Continuous random variables

Continuous RVs

take on uncountably many values



Given the probability space $(\Omega, \mathcal{A}, P)$, a continuous random variable is a function $X : \Omega \rightarrow \mathbb{R}$ such that $\forall x \in \mathbb{R}$ $\{\omega : X(\omega) \leq x\} \in \mathcal{A}$

Fully described by:
Cumulative Distribution Function
(CDF)
Probability Density Function
(PDF)

- PDF
- CDF
- Expectation
- Variance

Probability Density Function

Let X be a continuous random variable. The PDF of X is a function $f_X : \mathbb{R} \rightarrow \mathbb{R}$ given by

$$f_X(x) = \lim_{h \rightarrow 0} \frac{P(x < X \leq x + h)}{h} \quad \forall x \in \mathbb{R}$$

Properties:

- $f_X(x) \geq 0, \forall x \in \mathbb{R}$
- $\int_{-\infty}^{+\infty} f_X(x) dx = 1$
- $P(X = x) = 0, \forall x \in \mathbb{R}$
- $P(a \leq X \leq b) = \int_a^b f_X(x) dx$, for $a, b \in \mathbb{R}$ and $a < b$

Cumulative Distribution Function

Let X be a continuous random variable. The CDF of X is a function $F_X : \mathbb{R} \rightarrow \mathbb{R}$ given by

$$F_X(x) = P(X \leq x) \quad \forall x \in \mathbb{R}$$

Same properties as the CDF of discrete random variables:

- $0 \leq F_X(x) \leq 1$
- $\lim_{x \rightarrow -\infty} F_X(x) = 0, \quad \lim_{x \rightarrow +\infty} F_X(x) = 1$
- $F_X(x)$ is non-decreasing

Link between PDF and CDF

How are the PDF and CDF related?

- The CDF is the area under the PDF between $-\infty$ and x

$$F_X(x) = \int_{-\infty}^x f_X(t) dt$$

- The PDF is the derivative of $F_X(x)$, as

$$f_X(x) = \lim_{h \rightarrow 0} \frac{P(x < X \leq x + h)}{h} = \lim_{h \rightarrow 0} \frac{F_X(x + h) - F_X(x)}{h} = F'_X(x)$$

Exercise

Consider a continuous random variable X with the following PDF:

$$f_X(x) = \begin{cases} \frac{x}{8} & \text{for } 0 \leq x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

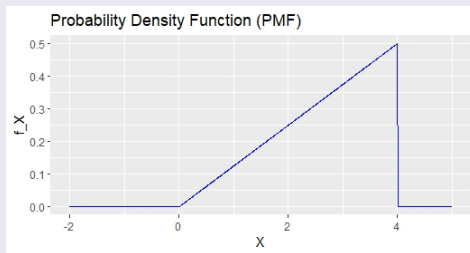
- 1 Plot the PDF
- 2 Calculate and plot the CDF, verify that it is a valid CDF
- 3 Determine $P(1 \leq X \leq 2)$ using the CDF

Exercise

Consider a continuous random variable X with the following PDF:

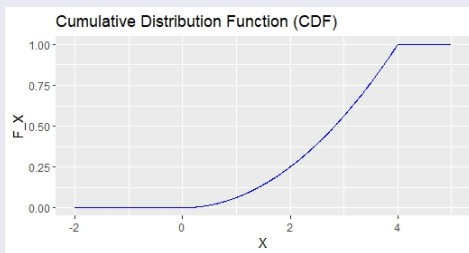
$$f_X(x) = \begin{cases} \frac{x}{8} & \text{for } 0 \leq x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

- 1 Plot the PDF
- 2 Calculate and plot the CDF, verify that it is a valid CDF
- 3 Determine $P(1 \leq X \leq 2)$ using the CDF



Exercise

- 2 The CDF is given by $F_X = \int_0^x \frac{t}{8} dt = \frac{x^2}{16}$
- ✓ $F_X(x) \geq 0, \forall x$
 - ✓ $\lim_{x \rightarrow -\infty} F_X(x) = 0, \lim_{x \rightarrow +\infty} F_X(x) = 1$
 - ✓ $F_X(x)$ is non-decreasing, as its first derivative $F'_X(x) = f_X(x)$ is positive for all $x \geq 0$
- 3 $P(1 \leq X \leq 2) = F(2) - F(1) = 4/16 - 1/16 = 3/16$



Expectation

Let X be a continuous random variable whose PDF is given by $f_X(x)$. The *expectation* or *mean* of X is given by

$$E[X] = \mu = \int_{-\infty}^{+\infty} x \cdot f_X(x) dx \quad (5)$$

More generally, we can define expectations for functions of random variables.

The expectation of a real-valued function $g(X)$ of a continuous random variable X with PDF given by $f_X(x)$, is

$$E[g(X)] = \int_{-\infty}^{+\infty} g(x) \cdot f_X(x) dx \quad (6)$$

Expectation

Special functions $g(X)$:

$g(X)$	$E[g(X)]$	
$g(X) = X$	$E[g(X)] = \int_{-\infty}^{+\infty} xf_X(x)dx = \mu$	Equation (6) reduces to (5), the mean of X
$g(X) = X^2$	$E[g(X)] = \int_{-\infty}^{+\infty} x^2 f_X(x)dx = \mu_2$	Second moment of X
$g(X) = (X - \mu)^2$	$E[g(X)] =$ $\int_{-\infty}^{+\infty} (x - \mu)^2 f_X(x)dx = \sigma^2$	Second centered moment of X or variance

Variance

Let X be a continuous random variable with mean μ and PDF $f_X(x)$.
The *variance* of X is given by

$$V[X] = \sigma^2 = E[(X - \mu)^2] = \int_{-\infty}^{+\infty} (x - \mu)^2 f_X(x) dx \quad (7)$$

Remark:

- Interpretation and properties of expectations are the same as in the discrete case

Exercise

Consider the previous exercise. Compute for the continuous variable X , the expected value $E[X]$, the second moment $E[X^2]$ and the variance $V[X]$

- Expected value

$$E[X] = \int_{-\infty}^{+\infty} x f_X(x) dx = \int_0^4 \frac{x^2}{8} dx = \left[\frac{x^3}{24} \right]_0^4 = \frac{8}{3}$$

- Second moment

Exercise

Consider the previous exercise. Compute for the continuous variable X , the expected value $E[X]$, the second moment $E[X^2]$ and the variance $V[X]$

- Expected value

$$E[X] = \int_{-\infty}^{+\infty} x f_X(x) dx = \int_0^4 \frac{x^2}{8} dx = \left[\frac{x^3}{24} \right]_0^4 = \frac{8}{3}$$

- Second moment

$$E[X^2] = \int_{-\infty}^{+\infty} x^2 f_X(x) dx = \int_0^4 \frac{x^3}{8} dx = \left[\frac{x^4}{32} \right]_0^4 = 8$$

- Variance

Exercise

Consider the previous exercise. Compute for the continuous variable X , the expected value $E[X]$, the second moment $E[X^2]$ and the variance $V[X]$

- Expected value

$$E[X] = \int_{-\infty}^{+\infty} x f_X(x) dx = \int_0^4 \frac{x^2}{8} dx = \left[\frac{x^3}{24} \right]_0^4 = \frac{8}{3}$$

- Second moment

$$E[X^2] = \int_{-\infty}^{+\infty} x^2 f_X(x) dx = \int_0^4 \frac{x^3}{8} dx = \left[\frac{x^4}{32} \right]_0^4 = 8$$

- Variance

$$V[X] = E[X^2] - E[X]^2 = 8 - \frac{64}{9} = \frac{8}{9}$$